



YouTube India Data Analysis

Project Overview: This project is an end-to-end data analysis case study focusing on the Indian YouTube market, explores trending video data to uncover patterns that influence video virality and channel growth strategies. Using Python, Pandas and Matplotlib in Jupyter Notebook, the dataset was structured and analyzed to study trend behavior and publishing strategy.

Aimed to understand how factors like real-time content and seasonal publishing strategies influence same-day trending probability and overall channel performance.

Defining Problem Statement: The analysis focuses on two core business question defined to understand content strategy and algorithm behavior:

Problem Statement 1: Same-Day Trending Analysis

Identify the patterns behind videos that reach to the **Trending** list on the very same day they are uploaded.

Problem Statement 2: Channel Analysis for T-Series

Analyze the upload strategy of India's biggest YouTube channel **T-series** to understand how they optimize their content schedule.

Assumptions: To frame the analysis, the following assumptions were made, to define hypothesis, regarding audience behavior and the YouTube algorithm:

- **Live Engagement Drives Trends:** We assume that live real-time content generate immediate viewership and high engagement rates in a short duration.
- **Seasonality Influences Demand:** We assume that the fluctuation in upload volume is not random but intentional, influenced by seasonal events.

Hypothesis 1: The Real-time Advantage Channel that frequently upload live streams have a significantly higher probability of their videos trending on the same day.

Hypothesis 2: The Seasonal Demand T-Series strategically upload content during months associated with high entertainment consumption, to capitalize on the festive mood of the

audience.

Exploratory Data Analysis:

🔗YouTube-Data-Analysis/C. YouTube Data Analysis.ipynb at main · shivangagrahari02-we...

Data Visualization:

🔗YouTube-Data-Analysis/D. Data Visualization.ipynb at main · shivangagrahari02-web/Yo...

Data Vizualization:

